

Strong Model Collapse

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Abstract

Within the scaling laws paradigm, which underpins the training of large neural networks like ChatGPT and Llama, we consider a supervised regression setting and establish the existence of a strong form of the model collapse phenomenon, a critical performance degradation due to synthetic data in the training corpus. Our results show that even the smallest fraction of synthetic data (e.g., as little as 1% of the total training dataset) can still lead to model collapse: larger and larger training sets do not enhance performance. We further investigate whether increasing model size, an approach aligned with current trends in training large language models, exacerbates or mitigates model collapse. In a simplified regime where neural networks are approximated via random projections of tunable size, we both theoretically and empirically show that larger models can amplify model collapse. Interestingly, our theory also indicates that, beyond the interpolation threshold (which can be extremely high for very large datasets), larger models may mitigate the collapse, although they do not entirely prevent it. Our theoretical findings are empirically verified through experiments on language models and feed-forward neural networks for images.

1 Introduction

The term Model Collapse refers to a critical degradation in the performance of AI models, particularly when a significant portion of their training data consists of synthetic data generated by other models. As detailed in Shumailov et al. (2023), this phenomenon arises as the model gradually overfits to patterns found in synthetic data, which may not fully represent the richness or variability of real-world data. Over successive training cycles, this feedback loop results in the model reinforcing errors, biases, or oversimplifications from the synthetic data. Consequently, the model’s ability to generalize to real-world data is compromised, as it increasingly relies on the distorted distribution provided by prior AI generations rather than learning accurate representations of the real world.

This phenomenon was observed empirically (Hataya et al., 2023; Martínez et al., 2023a,b; Bohacek & Farid, 2023; Briesch et al., 2023; Guo et al., 2023) and described theoretically (Alemohammad et al., 2023; Bertrand et al., 2023; Dohmatob et al., 2024a,b). The connection to the breakdown of neural scaling laws (Kaplan et al., 2020) has been pointed out and analyzed in Dohmatob et al. (2024b): as data becomes more synthetic, larger training sets do not enhance performance.

The issue is especially concerning in large-scale AI systems like ChatGPT and Llama (Touvron et al., 2023; Dubey & et al., 2024), which rely heavily on vast amounts of training data to maintain their performance. If synthetic data is used in training these models, even in small quantities, the model can start producing “gibberish” or nonsensical outputs, contains misinformation, or reflect stereotypes. This is because the model effectively starts to amplify its own mistakes (Shumailov et al., 2024). This feedback loop results in a gradual loss of model fidelity, reducing its ability to generalize or adapt to new, unseen test environments.

1.1 Main Contributions

In this work, we establish a series of results which shed more light on model collapse, bringing the phenomenon closer to a solid theoretical foundation. We consider the following important questions:

- (Q1) *Is model collapse inevitable or can it be fixed by strategically mixing synthetic and real data?*
(Q2) *Are larger models more prone to model collapse than smaller ones?*

Our theoretical analysis focuses on the solvable setting of linear regression with and without random projections, with the latter serving as an approximation of neural networks by means of random feature maps (Maloney et al., 2022; Bach, 2023). Also, in accordance with the current “neural scaling laws” paradigm (Kaplan et al., 2020; Hoffmann et al., 2022) which underlies the training of LLMs, where models and dataset sizes become larger over time, we focus on the setup where the total amount of data (synthetic + real data) used for training grows arbitrarily.

Let us summarize our main findings.

Result #1: Strong Model Collapse. First, we establish a robust negative result which shows that model collapse generally persists even when mixing real and synthetic data, as long as the fraction of training data which is synthetic does not vanish (cf. Section 3.1 and 3.2). By synthetic data, we mean any training data from a distribution which deviates from the distribution of real data, i.e. data on which the test performance is evaluated. Thus, model collapse cannot generally be mitigated by simple adjustments such as data weighting (Jain et al., 2024; Ferbach et al., 2024) unless these strategies asymptotically remove all but a vanishing proportion of synthetic data from the training process (Section 5). Our results show that the findings of Shumailov et al. (2024); Alemohammad et al. (2023); Bertrand et al. (2023); Dohmatob et al. (2024a,c) are worse than anticipated, by considering the more realistic scenario where a mixture of synthetic and real data is used for training.

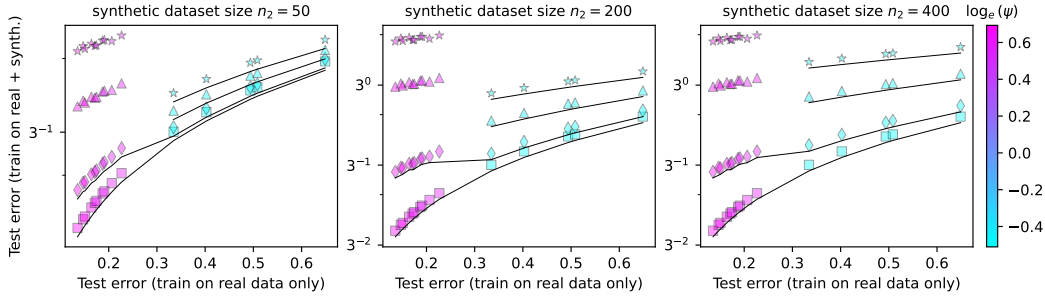


Figure 1: Pareto diagram: Understanding the role of model size in model collapse. We compare the test error (on the real / true data distribution), for a random projections model (equation (5) of Section 2.2) when training is done on a mix of synthetic and real data (y-axis), versus real data only (x-axis); in both cases, the total amount of training data is fixed to $n = 500$. On the scatter plots, square points correspond to very high-quality synthetic data (i.e. from a distribution which is close to the true data distribution), diamonds correspond to high-quality synthetic data, triangles correspond to low-quality, while stars correspond to very low-quality synthetic data. The black lines correspond to the Pareto frontiers for each level of quality of the synthetic data; the higher the frontier above the diagonal in the given setting, the more serious is the model collapse. The colorbar is the log of parametrization rate $\psi = m/n$, where m captures is the size of the model.

– **Result #2: Model Size and Model Collapse.** In Section 3.2, we disentangle the effect of a model’s size on its ability to cope with model collapse. We show that in general, bigger models will suffer more from model collapse as soon as the deviation between the distribution of the synthetic data and real data is significant. Crucially, our theory also predicts that past the interpolation threshold point, this tendency can be reversed: large models become more robust to model collapse. Put together, these results predict the existence of a double-descent curve regarding the model collapse phenomenon. This is illustrated in Figures 1 and 2. Thus, the model collapse profile depends critically on design choices like model size.

Experimental Validation. Our theoretical results are empirically confirmed with experiments in :

- Toy settings, including random projections model on Gaussian data, and shallow networks fully trained on the MNIST dataset (Deng, 2012). Refer to the end of Section 3.2 and Appendix ??.

- Realistic setting of GPT-2 models trained on BabiStories (Zhang et al., 2024a), a reproduction of TinyStories (Eldan & Li, 2023) using the Mixtral-8x7B open language model (Jiang et al., 2024)). Refer to Section 4.

Approach. From a technical standpoint, our theoretical analysis focuses on regression problems in the classical linear setting introduced in Dohmatob et al. (2024a) for studying model collapse, and also the setting of neural networks in a simplified regime which can be approximated by random projections (Maloney et al., 2022; Bach, 2023). We employ the tools of operator-valued free probability theory (OVFPT) (Mingo & Speicher, 2017) to obtain a new bias-variance decomposition $E_{test} \simeq B + V + \zeta$, of the test error evaluated on the real / true data distribution, of a model trained on a mixture of real and synthetic data. The extra term ζ then induces model collapse.

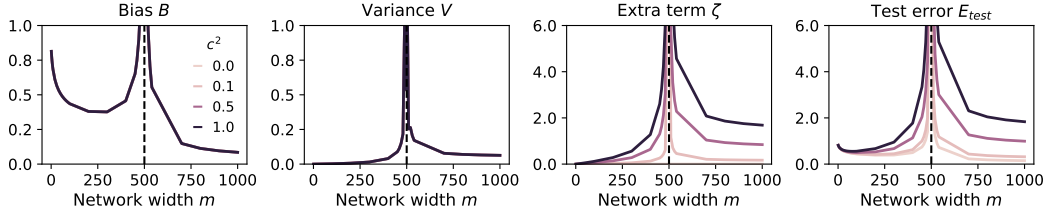


Figure 2: **Illustration of our new bias-variance decomposition** $E_{test} \simeq B + V + \zeta$ for neural networks in the simplified random projections regime (cf. Section 3.2), trained on a mixture of real and synthetic data. The sum $B + V$ corresponds to the classical bias variance decomposition in this setup when all the training data is real. The extra term ζ is responsible for model collapse when training is done on a mixture of real and synthetic data. The scalar c^2 characterizes the quality of the synthetic data (cf. Definition 1), via its mismatch with the real data distribution. The vertical line corresponds to the interpolation threshold $m = n$, where m is the model size and n is the total sample size. Notice the well-known double-descent curve in the bias curve.

1.2 Related Work

The theoretical study of model collapse in the setting of high-dimensional supervised-learning with linear regression and kernel ridge regression was initiated in Dohmatob et al. (2024a). This work derives analytical formulas that quantitatively describe iterative retraining on synthetic data in both under-parameterized and over-parameterized regimes, considering both low- and high-dimensional asymptotics. It places itself within an important body of works studying kernel ridge regression (on “clean” data), which serves as an effective proxy for neural networks in various regimes, for instance in the infinite-width limit (Neal, 1996; Williams, 1996; Jacot et al., 2018; Lee et al., 2018) or in the lazy regime of training (Chizat et al., 2019) and are a testbed to study interesting phenomena observed in deep learning. For instance, Rahimi & Recht (2008); Rudi & Rosasco (2017); Maloney et al. (2022) study scaling laws for regression in the random feature model and Bach (2023) analyses double descent in this setting. Scaling laws have been shown for kernel models under the Gaussian design, e.g. in Caponnetto & de Vito (2007); Spigler et al. (2020); Cui et al. (2022) for regression and Cui et al. (2023) for classification.

Very few theoretical works tackle the analysis of models trained on mixtures of original (real / clean) and synthetic data. Bertrand et al. (2023) analyze the training process at the distribution level and provide stability results under a locality assumption in parameter space. Seddik et al. (2024) analyze the mixing of discrete original and synthetic data, and provide upper bounds on the amount of synthetic data that can be included to avoid model collapse. Let us also mention the recent works (Jain et al., 2024; Ferbach et al., 2024) which are potential methods for mitigating model collapse. Jain et al. (2024) analyze linear regression on *isotropic* Gaussian data for mixtures of clean and synthetic data by minimizing a strategically weighted sum of losses (one term for each data source, real and synthetic), while Ferbach et al. (2024) can be seen as a multi-step version thereof where at each stage, the synthetic data generator is distilled by interpolating with real data. These methods are analyzed in Section 5, where we outline their shortcomings regarding model collapse.

Finally, a few works go beyond the mixing scenario and analyze how to curate or filter synthetic data to avoid model collapse (Feng et al., 2024; Zhang et al., 2024b; Alemohammad et al., 2024), but a rigorous study of their effectiveness is still lacking.